Propensity Score Workshop

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Preliminaries

Teaching Team Introductions

- Timothy (Tim) Feeney, UNC Chapel Hill, USA
- Catherine (Catie) Wiener, UNC Chapel Hill, USA
- Tobias (Tobias) Kurth, Charité, Berlin Germany

Outline of the workshop

1. Didactics

- Background and theory of basic propensity scores
- Estimands discussion
- Theory of inverse probability weighting
- Theory of standard error (SE) estimation

Outline of the workshop Con't

2. Practical Activities

- Breakout sessions to practice and use what we are teaching
- Will do propensity score analysis, boostrapping, m-estimation
- Will use R for coding. RMarkdown Files have been provided for reference

Schedule for the Afternoon

First half

1:10-1:20 Introduction to Clinical Problem

1:20-1:40 First Activity: Descriptive and Crude Analysis

1:40-2:40 Didactics: Overview of causal, estimands, propensity score basics

2:40-3:10 Second Activity: Estimation of ATE and ATT using IPTW

3:10-3:30 Break

Schedule for the Afternoon

Second half

3:30-4:10 Didactics: Standard Error estimation

4:10-4:50 Third Activity: Using Bootstrap and M-estimation

4:50-5:00 Questions and Closing Remarks

Acknowledgments

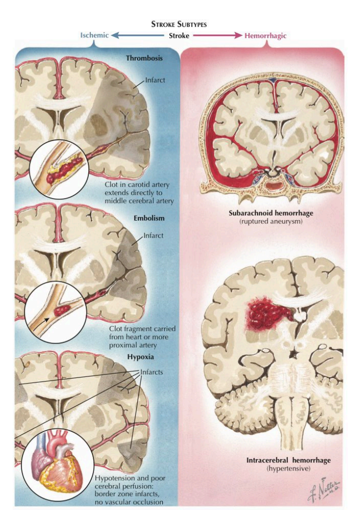
This workshop uses information from the following resources

- HSPH EPI 271 course materials
- UNC CH BIOS 776 Causal Notes
- What If Casual Textbook by *Hernan and Robins*
- Manuscripts by Kurth *et al.* 2006, Wiener *et al* 2025 and Ross *et al.* 2023
- Essential Statistical Inference by *Stefanski and Boos*

What do you want to get out of this course?

Question of Interest and Background

The Clinical Setup



Two General Types Of Stroke

The Clinical Setup

- Tissue plasminogen activatory (t-PA) can breakdown clots and has become a standard of care
- Kurth *et al.* in 2006 found that the effect of TPA in an observational cohort depends on method of confounding control, and when standardized to the whole sample tPA was harmful.
- PS distribution also suggested that there specific populations where there might be benefit

Dataset

Westphalian Stroke Registry of Northwestern Germany (Qualitätssicherung Schlaganfall Nordwestdeutschland, QSNWD)

- Data obtained from clinical records since 1999
- Contains information on:
  - sociodemographics, cerebrovascular factors, comorbidites
  - details of hospital, admission modality
  - stroke type and severity, procedures, complications, and status at discharge

FIRST ACTIVITY

FIRST ACTIVITY

Descriptive Statistics

Persuing Causal Estimates

The question of interest

What effect does tPA have on in-hospital all-cause mortality?

Is the effect the same in everyone? Is it the same in the whole population?

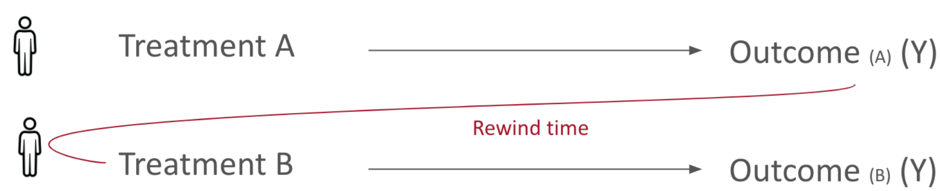
Approaches to finding an answer

	Description	Prediction	Causal inference
Example of scientific question	How can women aged 60–80 years with stroke history be partitioned in classes defined by their characteristics?	What is the probability of having a stroke next year for women with certain characteristics?	Will starting a statin reduce, on average, the risk of stroke in women with certain characteristics?
Data	• Eligibility criteria • Features (symptoms, clinical parameters …)	• Eligibility criteria • Output (diagnosis of stroke over the next year) • Inputs (age, blood pressure, history of stroke, diabetes at baseline)	• Eligibility criteria • Outcome (diagnosis of stroke over the next year) • Treatment (initiation of statins at baseline) • Confounders • Effect modifiers (optional)
Examples of analytics	Cluster analysis …	Regression Decision trees Random forests Support vector machines Neural networks …	Regression Matching Inverse probability weighting G-formula G-estimation Instrumental variable estimation …

Learning about causes of effects

- if we gave a person tPA and observed their outcome
- if instead we witheld tPA and observed the outcome

Would the outcomes be the same?



Learning about causes of effects

- This is the idea behind counterfactuals
- Let i index a person
- Let A_i and Y_i be RVs for treatment and the outcome
- $Y^{A=a}$ is the outcome had treatment $A = a$
- the comparisons of interest are $Y_i^{a=\text{tPA}} - Y_i^{a=\text{no tPA}}\ \forall i$
- in words: what would the individual effect be had each patient i recieved tPA versus not

Learning about causes of effects

One approach is to use a time machine

But we know this is not feasible (currently)

Tardis

Learning about causes of effects

- we cannot learn about individual effects;
  - we only have $Y_i^{a=\text{tPA}}$ OR $Y_i^{a=\text{no tPA}}$ for each i
  - no time machine or TARDIS
- Thus we calculate $\mathrm{E}\left[Y_i^{a=\text{tPA}} - Y_i^{a=\text{no tPA}}\right]$
- AKA the average treatment effect (ATE)
- The average effect had everyone gotten treatment, tPA, versus had everyone gotten no treatment, no tPA.

Estimating the Average Treatment Effect

Two ways to approach this:

	Benefits	**Problems**
Randomized Trials	Balanced groups in expectation	Perhaps infeasible, non generalizable
Observational Studies	Generalizable, potentially more feasible	Confounding, *et al*

We will focus on Observational Studies here

Causal Identification Conditions

Conditional Exchangeability: $Y^{a=1}, Y^{a=0} \perp\!\!\!\perp A|L$

Consistency: $Y = Y^{a=1}A + Y^{a=0}(1 - A)$

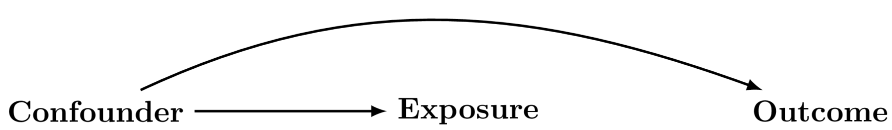
Positivity: $P[A = a|\mathbf{L}] > 0 \ \forall \ l$ where $f(l) > 0$

A major concern of non-randomized studies

Conditional Exchangeability: $Y^{a=1}, Y^{a=0} \perp\!\!\!\perp A|L$

Confounders are a major concern

Confounder: On a causal path to the outcome AND on a causal path to the exposure AND not caused by the the outcome or exposure.



Methods to control for confounding

- Matching
- Restriction
- Regression Modeling
- Weighting

These are all amenable to propensity methods

We will focus on weighting

Propensity Scores

Propensity Score- Definition

The propensity score is the probability of being in a treatment group conditional on covariates–it is the propensity to be treated

$$P[A = 1|\mathbf{L}]$$

where A is treatment level and $\mathbf{L}$ is a vector of covariates.

We will denote propensity score as $\pi(\mathbf{L})$ for simplicity

Propensity score- Exhangeability

The propensity score is a balancing score[1]. If we have a set of $\mathbf{L}$ such that $\left(Y^{a=0}, Y^{a=1}\right) \perp\!\!\!\perp A \mid \mathbf{L}$ then

$$\left(Y^{a=0}, Y^{a=1}\right) \perp\!\!\!\perp A \mid \pi\left(\mathbf{L}\right)$$

In other words, if $\mathbf{L}$ represents a vector that contains all variables necessary to achieve exchangeability then the propensity score can provide exchangeability.

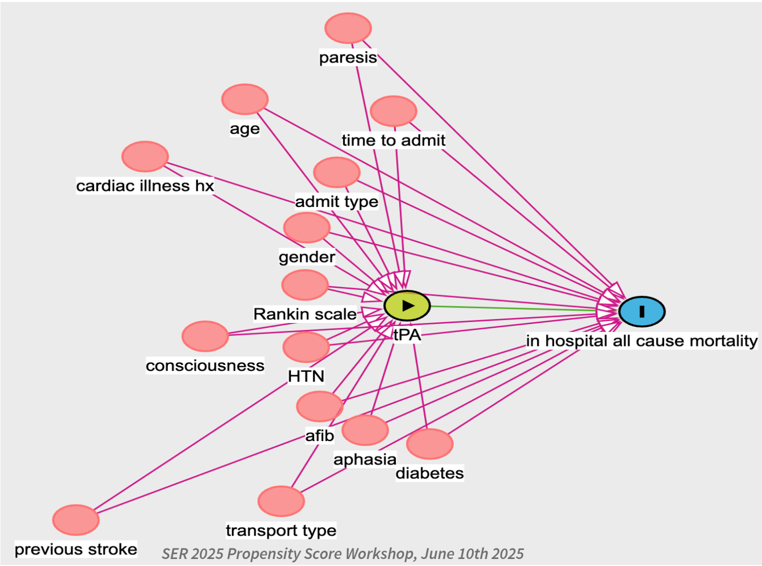
How do we determine what should be in $\mathbf{L}$?

Model building and variable selection

Like all causal analysis, variable selection should be principled, guided by your estimand, and driven by your DAG

1. Identify Variables in your DAG that you believe may be confounders.[1]
2. Exclude variables that are
    - a. mediators of your effect of interest
    - b. instruments

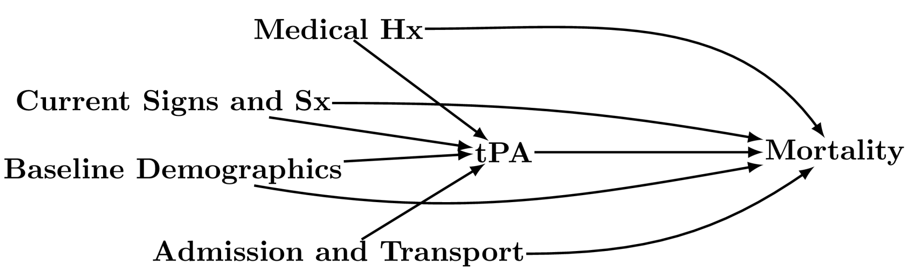
Build the DAG



Collapse Categories in the DAG

Medical Hx	Sx and Signs	Demographics	Admission
Cardiac Illness Hx	Paresis	Age	Transport Type
HTN	Aphasia	Gender	Time To admission
AFib	Consciousness	Admission Location	
Diabetes	Rankin Scale		

Representative DAG



Propensity Score Estimation

Typically use logistic regression to estimate.

$$\text{logit}\left[\hat{\pi}\left(\mathbf{L}\right)\right] = \text{logit}\left(P\left[A = 1 \mid \mathbf{L}\right]\right) = \alpha_0 + \boldsymbol{\alpha}^T\mathbf{L}$$

and

$$\hat{\pi}\left(\mathbf{L}\right) = \text{expit}\left[\text{logit}\left(P\left[A = 1 \mid \mathbf{L}\right]\right)\right] = \text{expit}\left[\alpha_0 + \boldsymbol{\alpha}^T\mathbf{L}\right]$$

where $\text{logit} = \log\left(\frac{P[A=1|\mathbf{L}]}{1-P[A=1|\mathbf{L}]}\right)$ and $\text{expit} = \left(\frac{1}{1+\exp^{(\alpha_0+\boldsymbol{\alpha}^T\mathbf{L})}}\right)$

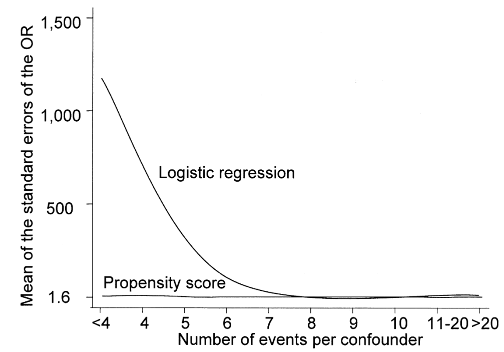
in R

```r
#model the logit of being in treatment
model<-glm(A~L, data=df, family=binomial(link='logit'))

#predict the probability of being in treatment based on covariates
prop_score<-predict(model, type='response')
```

Advantages

- Collapses predictors of treatment into a single value.
  - avoid curse of dimensionality.[1]
- Relatively easy to estimate.
- Multiple uses.



Disadvantages

- Still rely on the assumption that you have all variables necessary to achieve exchangeability.
  - **There is no magic here, this is not an RCT**
- People with same propensity score may have different distribution of covariates.
- Nuisance parameter estimation apdds an extra step to estimation.
- Estimation of standard error may be more complex.

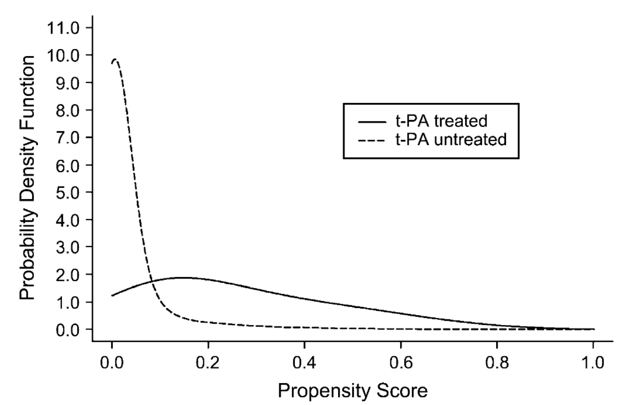
Propensity Score Diagnostics

-Distribution of scores

-Overlap

-Trimming?

-Truncation?



Inverse Probability Weighting

Inverse Probability of Treatment Weights

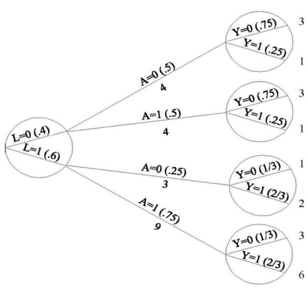
- In Epidemiology we are often interested in marginal estimates of effect
- To achieve this we often use Inverse Probability of Treatment weights (IPTW)

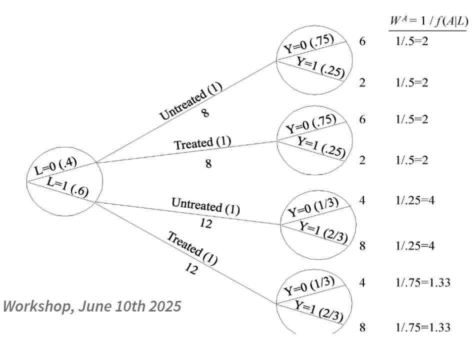
$$\text{IPTW} = W_{IPTW} = (f(A \mid \mathbf{L}))^{-1} = \frac{1}{P[A \mid \mathbf{L}]}$$

How IPTW works

Table 2.2

	L	A	Y
Rheia	0	0	0
Kronos	0	0	1
Demeter	0	0	0
Hades	0	0	0
Hestia	0	1	0
Poseidon	0	1	0
Hera	0	1	0
Zeus	0	1	1
Artemis	1	0	1
Apollo	1	0	1
Leto	1	0	0
Ares	1	1	1
Athena	1	1	1
Hephaestus	1	1	1
Aphrodite	1	1	1
Polyphemus	1	1	1
Persephone	1	1	1
Hermes	1	1	0
Hebe	1	1	0
Dionysus	1	1	0

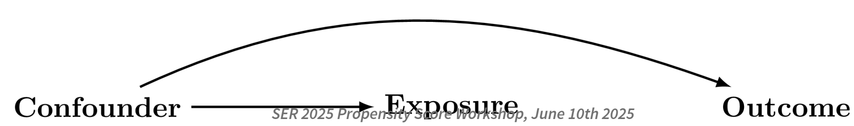




How IPTW works

Before

	$A = 1$	$A = 0$	
$L = 0$	4	4	
$L = 1$	9	3	
Total	13	7	20



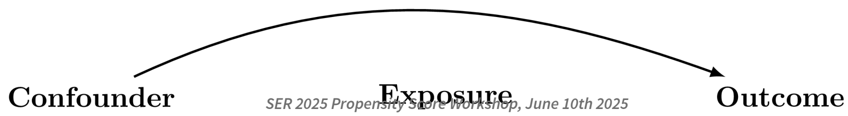
How IPTW works

Before

	$A = 1$	$A = 0$	
$L = 0$	4	4	
$L = 1$	9	3	
Total	13	7	20

After

	$A = 1$	$A = 0$	
$L = 0$	8	8	
$L = 1$	12	12	
Total	20	20	40



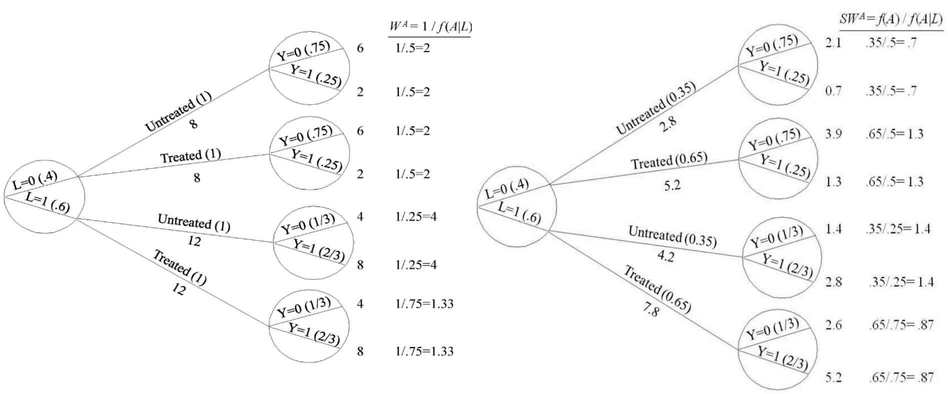
Stabilized Versus Unstabilized

$$\widehat{\mathrm{W}}_{US,i} = \frac{1}{f(A|\mathbf{L})} = \frac{1}{P[A|\mathbf{L}]}$$

Stabilized weights are scaled by the probability of treatment.

$$\widehat{\mathrm{W}}_{S,i} = \frac{f(A)}{f(A|\mathbf{L})} = \frac{P[A]}{P[A|\mathbf{L}]}$$

Stabilized Versus Unstabilized



Estimation Examples

Example ATE estimation - Setup

What is the ATE of quitting smoking versus not on weight change between 1971 and 1982?

Using NHEFS Data from Hernan and Robins

Treatment of interest: Quitting smoking (qsmk) Confounders: Sex, Age, Race

Description of NHEFS data

```
set.seed(12345) #set seed for todays
require(causaldata)
require(dplyr)
ex<-causaldata::nhefs %>% select(wt82_71, qsmk, age, race, sex) %>% filter(
ex
```

```
# A tibble: 6 × 5
  wt82_71  qsmk   age race  sex
    <dbl> <dbl> <dbl> <fct> <fct>
1  -10.1      0    42 1     0
2    2.60     0    36 0     0
3    9.41     0    56 1     1
4    4.99     0    68 1     0
5    4.99     0    40 0     0
6    4.42     0    43 1     1
```


Estimating the ATE, Horvitz Thompson Estimator

$$\frac{1}{n}\sum_i \hat{W}_i Y_i A_i - \frac{1}{n}\sum_i \hat{W}_i Y_i (1 - A_i)$$

Estimating the ATE, Horvitz Thompson Estimator Setup

```
require(dplyr)
nhefs<-causaldata::nhefs %>% filter(!is.na(wt82_71))
# Model treatment
trt_model<-glm(qsmk ~ as.factor(sex) + as.factor(race) + age + I(age^2), da
                  family=binomial())
# Estimate propensity score
nhefs$prop_score<-ifelse(nhefs$qsmk==1, predict(trt_model, type="response")
 1-predict(trt_model, type="response"))
# Estimate unstabilized IPTW
nhefs$w<-1/nhefs$prop_score
```

```
# A tibble: 4 × 7
  wt82_71  qsmk   age race  sex   prop_score     w
    <dbl> <dbl> <dbl> <fct> <fct>      <dbl> <dbl>
1  -10.1      0    42 1     0          0.814  1.23
2    2.60     0    36 0     0          0.742  1.35
3    9.41     0    56 1     1          0.807  1.24
4    4.99     0    68 1     0          0.698  1.43
```

Estimating the ATE, Horvitz Thompson Estimator Code

$$\frac{1}{n}\sum_i \hat{W}_i Y_i A_i - \frac{1}{n}\sum_i \hat{W}_i Y_i (1 - A_i)$$

```
nhefs$treat<-nhefs$w*nhefs$wt82_71*nhefs$qsmk
nhefs$untrt<-nhefs$w*nhefs$wt82_71*(1-nhefs$qsmk)
ate<-mean(nhefs$treat, na.rm=T)-mean(nhefs$untrt, na.rm=T)
ate
```

```
[1] 3.09156
```


Example ATE, Hajek Estimator

We may instead want to use regression using least squares

If instead of minimizing $\sum_i \left[Y_i - (\widehat{\alpha_0} + \widehat{\alpha_1} A_i)\right]^2$

we minimize

$$\sum_i \widehat{W_i}\left[Y_i - (\widehat{\alpha_0} + \widehat{\alpha_1} A_i)\right]^2$$

This is equivalent to using the Hajek estimator

$$\hat{\alpha}_1 = \frac{\sum_i \widehat{W_i} Y_i A_i}{\sum_i \widehat{W_i} A_i} - \frac{\sum_i \widehat{W_i} Y_i (1 - A_i)}{\sum_i \widehat{W_i} (1 - A_i)}$$

Estimating the ATE, Hajek Estimator Code

$$\frac{\sum_i \widehat{W}_i Y_i A_i}{\sum_i \widehat{W}_i A_i} - \frac{\sum_i \widehat{W}_i Y_i (1 - A_i)}{\sum_i \widehat{W}_i (1 - A_i)}$$

```
# weighting the regression model
Y_model_ate<-lm(wt82_71~qsmk, weights=w, data=nhefs)

# Horvitz Thompson ATE
paste0("ATE estimate using Hortvitz Thompson: ",round(ate,4))
```

```
[1] "ATE estimate using Hortvitz Thompson: 3.0916"
```

```
# regression based ATE (Hajek estimator)
paste0("ATE estimate using Hajek: ",round(Y_model_ate$coefficients[2],4))
```

```
[1] "ATE estimate using Hajek: 3.085"
```

Advantages of using Propensity Score Weights

- Can be used in many scenarios to obtain marginal estimates
  - Average Treatment Effect [1] -IPTW, done
  - Average Treatment Effect among Treated (ATT)/ Untreated (ATU) [2] - SMR/SRR, next
- Uses all the data

The question of interest

What effect does tPA have on in-hospital all-cause mortality?

Is the ATE what we are interested in?

Do we expect that everyone coming into an ED with stroke symptoms would benefit from tPA?

What if were were interested in the effect of giving everyone treatment versus not *only* in the treated?

What if were were interested in the effect of giving everyone treatment versus not *only* in the untreated?

ATT

$$E\left[Y^{a=1} - Y^{a=0} \middle| A = 1\right]$$

Average Effect among those that were treated.

Can be obtained using Standardized Mortality Ratio (SMR) weights[1]

Why might this be more pertinent?

Estimating the ATT

Under a slightly different set of identification conditions:

Consistency: $Y = Y^{a=1}A + Y^{a=0}(1 - A)$

Positivity: $P[A = a|\mathbf{L}] > 0 \ \forall \ l$ where $f(l|a = 1) > 0$

Conditional Exchangeability: $Y^{a=0} \perp\!\!\!\perp A|L$

Estimating the ATT

Positivity: $P[A = a|\mathbf{L}] > 0 \ \forall \ l$ where $f(l|a = 1) > 0$

Conditional Exchangeability: $Y^{a=0} \perp\!\!\!\perp A|L$

Estimating SMR weights for the ATT

Weights to make the untreated look like the treated

$$\widehat{W}_{i\ SMR} = \frac{P[A = 1 \mid \mathbf{L}]}{P[A = a \mid \mathbf{L}]} \implies \widehat{W}_{i\ SMR} = \begin{cases} a = 1, & 1 \\ a = 0, & \frac{\hat{\pi}(\mathbf{L})}{1-\hat{\pi}(\mathbf{L})} \end{cases}$$

```
require(dplyr); nhefs<-causaldata::nhefs
# Model treatment
trt_model<-glm(qsmk ~ as.factor(sex) + as.factor(race) + age + I(age^2), da
                  family=binomial())
# Estimate propensity score
nhefs$prop_score<-ifelse(nhefs$qsmk==1, 1,
 predict(trt_model, type="response")/(1-predict(trt_model, type="response")
 nhefs$w_att<-1/nhefs$prop_score
```

```
[1] "The estimated ATT is: 3.05"
```


ATU

$$E\left[Y^{a=1} - Y^{a=0} \middle| A = 0\right]$$

Average Effect among those that were untreated.

Can be obtained using Standardized Risk/Rate Ratio (SRR) weights[1]

Why might this not be as interesting in our case?

Estimating the ATU

Under a slightly different set of identification conditions:

Consistency: $Y = Y^{a=1}A + Y^{a=0}(1 - A)$

Positivity: $P[A = a|\mathbf{L}] > 0 \ \forall \ l$ where $f(l|a = 0) > 0$

Conditional Exchangeability: $Y^{a=1} \perp\!\!\!\perp A|L$

Estimating the ATU

Positivity: $P[A = a|\mathbf{L}] > 0 \;\forall\; l$ where $f(l|a = 0) > 0$

Conditional Exchangeability: $Y^{a=1} \perp\!\!\!\perp A|L$

Estimating SRR Weights for the ATU

New weights to make the treated look like the untreated, thereby standing in for them.

$$\widehat{W}_{i\,SRR} = \frac{P\left[A = 0 \mid \mathbf{L}\right]}{P\left[A = a \mid \mathbf{L}\right]} \implies \widehat{W}_{i\,SRR} = \begin{cases} a = 1, & \frac{1-\widehat{\pi}(\mathbf{L})}{\widehat{\pi}(\mathbf{L})} \\ a = 0, & 1 \end{cases}$$

```
require(dplyr);nhefs<-causaldata::nhefs
# Model treatment
trt_model<-glm(qsmk ~ as.factor(sex) + as.factor(race) + age + I(age^2), da
                  family=binomial())
# Estimate propensity score
nhefs$prop_score<-ifelse(nhefs$qsmk==0, 1,
 (1-predict(trt_model, type="response"))/predict(trt_model, type="response"
 nhefs$w_atu<-1/nhefs$prop_score
```

```
[1] "The estimated ATU is: 3.1"
```


SECOND ACTIVITY

SECOND ACTIVITY

Estimate the ATE, ATT, and ATU using IPTW in the tPA data

1. Explain in words what each estimand means in this context, and which estimand you think has meaning for our question of interest.
2. Estimate the ATE, ATT, and ATU in the data.

BREAK

Standard Errors

Standard Error Estimation:

- Until now we have discussed effect estimation but have not talked about Standard Error (SE) estimation.
- This is more complicated
  - Multiple estimation steps need to be accounted for
  - No closed form solution for this
  - Need alternative methods

Robust Variance Estimation

- Commonly cited method of obtaining SE estimates, referred to a 'Huber-White' SEs
- The Sandwich Variance Estimator

$$\hat{V} = (-A)^{-1} B(-A)^{-1}$$

- **Assumes weights are known.**
- Conservative in estimating ATE, Conservative or Anticonservative for the ATT[1]

Robust Variance Estimation

R-Code

```r
sandwich:::sandwich(Y_model_att)
```

or

```r
summary(geeglm(wt82_71~qsmk, weights=w_att, data=nhefs, id=seqn))
```

Standard Errors By Method

Method	ATE	ATT	ATU
Model	0.405	0.408	0.404
Robust	0.486	0.492	0.493

Standard Error Estimation- Bootstrapping

1. From a sample of $1 \ldots N$ individuals, sample a total of N individuals with replacement, resulting in new sample of size K, where $K = N$
2. Perform estimation on new sample of $1 \ldots K$ individuals
3. Repeat this process j times; $j \geq 200 - 300$ but ideally many more (e.g., 3000-5000). Also worry about machine error with low numbers [1].

Standard Error Estimation- Bootstrapping Cont.

4. Use the standard error or percentile cut-offs from the distribution of estimates to estimate the empirical standard error.
5. Use this empirical standard error to calculate Wald confidence intervals.

Standard Error Estimation-Bootstrapping

From Bootstrapping:

```
atu_boot<-boot(nhefs,5000, estimand="ATU")
```

```
[1] "The bootstrap ATU estimate is: 3.098"
[1] "The ATU Bootstrap SE is: 0.474"
```

From estimation above

```
[1] "The estimated ATU is: 3.1"
```


Standard Error Estimation-Bootstrapping Problems:

- Time

```
> system.time(atu_boot<-boot(nhefs,100, estimand= "ATU"))
   user  system elapsed
  0.250   0.022   0.273
> system.time(atu_boot<-boot(nhefs,1000, estimand= "ATU"))
   user  system elapsed
  2.675   0.242   2.917
> system.time(atu_boot<-boot(nhefs,10000, estimand= "ATU"))
   user  system elapsed
  26.91    2.44   29.52
```


Variance Estimation–Updated

Standard Errors By Method

Method	ATE	ATT	ATU
Model	0.405	0.408	0.404
Robust	0.486	0.492	0.493
Bootstrap	0.471	0.475	0.474

M-Estimation

NB: There is no way to cover all of M-Estimation in the remainder of this session but we will hopefully cover aspects for you to make use of it.

M-Estimation

Excellent sources to dive deeper:

- Ross *et al.* M-estimation for common epidemiological measures: introduction and applied examples IJE 2024
- Boos and Stefanski. Essential Statistical Inference 2013 Chapter 7
- An introduction to M-estimation with geex https://cran.r-project.org/web/packages/geex/vignettes/v00_geex_intro.html
- ABC's of M-Estimation SER Workshop (https://github.com/pzivich/ABCs_of_M-estimation/)

M-Estimation

M-estimation addresses all concerns above.

- computationally much more efficient
- consistent: estimates close to estimand as $n \longrightarrow \infty$
- asymptotically normal estimates: can use SEs in Wald type CIs
- allows you to incorporate prior estimation steps (i.e., estimation of weights)
  - does not require assuming the weights are known as in Sandwich estimator

M-Estimation

M-estimators, $\widehat{\boldsymbol{\theta}}$, are solutions to:

$$\sum_{i=1}^{n} \psi\left(\boldsymbol{Y}_i, \widehat{\boldsymbol{\theta}}\right) = \mathbf{0}.$$

where $Y_i, \ldots, Y_n$ are observed data and are independent but not necessarily identically distributed

$\boldsymbol{\theta}$ is a $p \times 1$ vector of parameters

ψ is a known $p \times 1$ function not dependent on n or i.

M-Estimation

We can stack estimating functions

$$\sum_{i=1}^{n} \begin{bmatrix} \psi_{\theta_1}\left(O_i; \widehat{\theta}\right) \\ \psi_{\theta_2}\left(O_i; \widehat{\theta}\right) \\ \vdots \\ \psi_{\theta_k}\left(O_i; \widehat{\theta}\right) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \mathbf{0}$$

The benefit is that during estimation, previous steps can be accounted for.

M-Estimation

Recall:

$$\hat{\theta} \text{ is AN} \left(\theta_0 , \frac{\boldsymbol{V}\left(\theta_0\right)}{n} \right) \text{ as } n \to \infty,$$

Where an estimator for $\boldsymbol{V}(\boldsymbol{\theta_0})$ is the empirical sandwich variance estimator:

$$\boldsymbol{V}_n(\boldsymbol{Y},\hat{\theta}) = \boldsymbol{A}_n(\boldsymbol{Y},\hat{\theta})^{-1}\, \boldsymbol{B}_n(\boldsymbol{Y},\hat{\theta})\left\{\boldsymbol{A}_n(\boldsymbol{Y},\hat{\theta})^{-1}\right\}^T.$$

M-Estimation Variance

A litte more on variance estimation

$$\boldsymbol{V}_n(\boldsymbol{Y},\hat{\theta}) = \boldsymbol{A}_n(\boldsymbol{Y},\hat{\theta})^{-1}\, \boldsymbol{B}_n(\boldsymbol{Y},\hat{\theta})\left\{\boldsymbol{A}_n(\boldsymbol{Y},\hat{\theta})^{-1}\right\}^T.$$

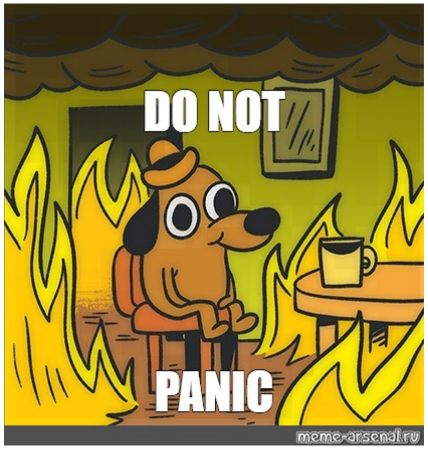
$$\boldsymbol{A_n}(\boldsymbol{Y},\hat{\theta}) = \frac{1}{n}\sum_{i=1}^{n}\left\{-\boldsymbol{\psi}'\left(Y_i,\hat{\theta}\right)\right\}$$

and

$$\boldsymbol{B}_n(\boldsymbol{Y},\hat{\theta}) = \frac{1}{n}\sum_{i=1}^{n}\boldsymbol{\psi}\left(Y_i,\hat{\theta}\right)\boldsymbol{\psi}\left(Y_i,\hat{\theta}\right)^T$$

M-Estimation

This might be overwhelming, so lets work through an example to illustrate this.



M-Estimation-Example

Lets estimate the mean and variance using M-Estimation

$$\widehat{\mu} = \frac{1}{n} \sum_{i=1}^{n} Y_i$$

Where μ is the mean, $\hat{\mu}$ is an estimate of the mean and Y_i is the ith persons outcome

How do we get this into an estimating equation?

Recall, we are trying to get into the form: $\sum_{i=1}^{n} \psi(\boldsymbol{Y}_i, \boldsymbol{\theta}) = \mathbf{0}$

M-Estimation-Example

$$\widehat{\mu} n = \sum_{i=1}^{n} Y_i$$

$$0 = \left(\sum_{i=1}^{n} Y_i\right) - \widehat{\mu} n$$

$$0 = \left(\sum_{i=1}^{n} Y_i\right) - \left(\sum_{i=1}^{n} \widehat{\mu}\right)$$

$$0 = \sum_{i=1}^{n} (Y_i - \widehat{\mu})$$

M-Estimation-Example

A similar process for variance gives us:

$$\boldsymbol{\psi}(Y_i, \boldsymbol{\theta}) = \begin{pmatrix} Y_i - \theta_1 \\ (Y_i - \theta_1)^2 - \theta_2 \end{pmatrix}.$$

Our Estimating equation is:

$$\sum_{i=1}^{n} \boldsymbol{\psi}\left(Y_i, \widehat{\boldsymbol{\theta}}\right) = \begin{pmatrix} \sum_{i=1}^{n} \left(Y_i - \widehat{\theta}_1\right) \\ \sum_{i=1}^{n} \left((Y_i - \widehat{\theta}_1)^2 - \widehat{\theta}_2\right) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The M-estimator is $\widehat{\boldsymbol{\theta}} = \left(\bar{Y}, s_n^2\right)^T$.

M-Estimation-Example

Now we need to make a computer do this for us.

First Set up the necessary packages and create some example data

```r
library("numDeriv")
library("rootSolve")
library("geex")

y <- geexex$Y1  # Data (observations)
n <- length(y)      # Number of observations
```


M-Estimation-Example

Next set up the equations

```r
estimating_function <- function(mu){
  mean = y - mu[1]
  var=(y-mu[1])^2-mu[2]
  return(cbind(mean,var))
}

estimating_equation <- function(mu){
  estf = estimating_function(mu)
  este = colSums(estf)
  return(este)
}
```


M-Estimation-Example

Next use root finding to obtain estimates

```r
# Root-finding

mu <- rootSolve::multiroot(f = estimating_equation,
                           start = c(0,0))
mu_root <- mu$root
```

```
[1] "Root finding estimate of mean: 5.045"
```

Compare this to the close form solution using `sum(y)/n` in R

```
[1] "Closed form estimate of mean: 5.045"
```

M-Estimation-Example

Next we need to build the components of

$$V_n(Y,\hat{\theta}) = A_n(Y,\hat{\theta})^{-1} B_n(Y,\hat{\theta}) \left\{ A_n(Y,\hat{\theta})^{-1} \right\}^T.$$

M-Estimation-Example

Lets start with

$$A_n(Y,\hat{\theta}) = \frac{1}{n}\sum_{i=1}^{n}\left\{-\psi'\left(Y_i,\hat{\theta}\right)\right\}$$

```
deriv <- numDeriv::jacobian(func = estimating_equation,   # Function to fir
                            x = mu_root)                  # Compute Derivat

A <- -1*deriv / n
print(A)
```

```
              [,1] [,2]
[1,]  1.000000e+00    0
[2,] -4.032627e-12    1
```


M-Estimation-Example

Next

$$B_n(Y,\hat{\theta}) = \frac{1}{n}\sum_{i=1}^{n}\psi\left(Y_i,\hat{\theta}\right)\psi\left(Y_i,\hat{\theta}\right)^T$$

```
outerprod <- t(estimating_function(mu_root)) %*% (estimating_function(mu_rc

B <- outerprod/n
print(B)
```

```
          mean        var
mean 10.041239   3.667969
var   3.667969 249.219638
```


M-Estimation-Example

combine the components:

$$V_n(Y,\hat{\theta}) = A_n(Y,\hat{\theta})^{-1} B_n(Y,\hat{\theta}) \left\{ A_n(Y,\hat{\theta})^{-1} \right\}^T.$$

```
sandwich <- solve(A) %*% B %*% t(solve(A))
print(sandwich)
```

```
          [,1]       [,2]
[1,] 10.041239   3.667969
[2,]  3.667969 249.219638
```

The sample variance is 10.0412

Closed form solution using `sum((y-sum(y)/n)^2)/n` = 10.0412

M-Estimation-Example

But to get the SE recall

$$\hat{\theta} \text{ is AN}\left(\theta_0, \frac{V(\theta_0)}{n}\right) \text{ as } n \to \infty,$$

```r
se<-sqrt(diag(sandwich)/n)
print(se)
```

```
[1] 0.3168791 1.5786692
```


M-Estimation with geex in R

Setup the function

```r
mean_est<-function(data){
     y<-data$Y1
     function(theta){
     c(y - theta[1],          #mean
     (y-theta[1])^2-theta[2]) #var
  }
}
```


M-Estimation with geex in R

Feed to geex

```r
results <- m_estimate(
    estFUN = mean_est,
    data   = geexex,
    root_control = setup_root_control(start = c(1,1)))

print(results@estimates)
```

```
[1]  5.044563 10.041239
```

```r
print(results@vcov)
```

```
           [,1]       [,2]
[1,] 0.10041239 0.03667969
[2,] 0.03667969 2.49219638
```

```r
print(sqrt(diag(vcov(results))))
```

```
[1] 0.3168791 1.5786692
```


M-estimation for IPW ATE and ATT

Build estimating functions for estimating the ATE and ATT.

First, set up functions to estimate the propensity score

$$\psi\left(Y_i, \hat{\theta}\right) = \begin{bmatrix} A_i - \text{expit}\left(\hat{\beta}_0 + \hat{\beta}_1 W_i\right) \\ \left(A_i - \text{expit}\left(\hat{\beta}_0 + \hat{\beta}_1 W_i\right)\right) W_i \end{bmatrix}$$

Where

$$\hat{\pi}_{IPT,i} = \frac{A_i}{\text{expit}\left(\hat{\beta}_0 + \hat{\beta}_1 W_i\right)} + \frac{1 - A_i}{1 - \text{expit}\left(\hat{\beta}_0 + \hat{\beta}_1 W_i\right)}$$

and

$$\hat{\pi}_{SMR,i} = A_i + \frac{(1 - A_i) * \text{expit}\left(\hat{\beta}_0 + \hat{\beta}_1 W_i\right)}{1 - \text{expit}\left(\hat{\beta}_0 + \hat{\beta}_1 W_i\right)}$$

M-estimation for IPW ATE and ATT

Next setup the function for estimating the weighted control and treatment group means

$$\psi\left(Y_i, \hat{\boldsymbol{\theta}}\right) = \begin{bmatrix} A_i Y_i \hat{\pi}_{IPT,i} - \hat{\theta}_3 \\ (1 - A_i)\, Y_i \hat{\pi}_{IPT,i} - \hat{\theta}_4 \\ A_i \hat{\pi}_{IPT,i} \left(Y_i - \hat{\theta}_5\right) \\ (1 - A_i)\, \hat{\pi}_{IPT,i} \left(Y_i - \hat{\theta}_6\right) \\ A_i \hat{\pi}_{SMR,i} \left(Y_i - \hat{\theta}_7\right) \\ (1 - A_i)\, \hat{\pi}_{SMR,i} \left(Y_i - \hat{\theta}_8\right) \end{bmatrix}$$

M-estimation for IPW ATE and ATT

Last estimate the ATE and ATT

$$\psi\left(Y_i, \hat{\boldsymbol{\theta}}\right) = \begin{bmatrix} \hat{\theta}_5 - \hat{\theta}_6 - \hat{\theta}_9 \\ \hat{\theta}_7 - \hat{\theta}_8 - \hat{\theta}_{10} \end{bmatrix}$$

M-estimation for IPW ATE and ATT-Combined

$$\psi\left(Y_i, \hat{\boldsymbol{\theta}}\right) = \begin{bmatrix} A_i - \operatorname{expit}\left(\hat{\beta}_0 + \hat{\beta}_1 W_i\right) \\ \left(A_i - \operatorname{expit}\left(\hat{\beta_0} + \hat{\beta}_1 W_i\right)\right) W_i \\ A_i Y_i \hat{\pi}_{IPT,i} - \hat{\theta}_3 \\ (1 - A_i)\, Y_i \hat{\pi}_{IPT,i} - \hat{\theta}_4 \\ A_i \hat{\pi}_{IPT,i} \left(Y_i - \hat{\theta}_5\right) \\ (1 - A_i)\, \hat{\pi}_{IPT,i} \left(Y_i - \hat{\theta}_6\right) \\ A_i \hat{\pi}_{SMR,i} \left(Y_i - \hat{\theta}_7\right) \\ (1 - A_i)\, \hat{\pi}_{SMR,i} \left(Y_i - \hat{\theta}_8\right) \\ \hat{\theta}_5 - \hat{\theta}_6 - \hat{\theta}_9 \\ \hat{\theta}_7 - \hat{\theta}_8 - \hat{\theta}_{10} \end{bmatrix}$$

Variance Estimation–Update 2!

Standard Errors By Method

Method	ATE	ATT	ATU
Model	0.405	0.408	0.404
Robust	0.486	0.492	0.493
Bootstrap	0.471	0.475	0.474
M-Est	0.469	0.466	-

THIRD ACTIVITY

THIRD ACTIVITY

Estimate SEs using

- Robust Variance
- Boostrap
- M-estimation

Thank you

We appreciate you all for attending

Questions?